Project Design Phase-II
Data Flow Diagram & User Stories

Date	09 March 2026
Team ID	
Project Name	Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau
Maximum Marks	4 Marks

1. Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is used to visually represent how data moves through a system. It shows:

- Where data comes from
- How the data is processed
- Where the data is stored
- How results are delivered to users

In this project, the DFD explains how housing market data flows from datasets to Tableau dashboards and finally to users through a Flask web application.

2. DFD Level-0 (Context Diagram)

Explanation

The **Level-0 DFD** represents the entire system as a single process.

External Entities

1. Data Sources

The system collects housing data from:

- Kaggle housing dataset
- CSV or Excel files
- Real estate property databases

These datasets contain information such as:

- Sale price
- House age
- Bedrooms
- Bathrooms
- Floors
- Renovation year
- Basement area

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2. Housing Market Analysis System

The main system processes the data using:

- Data preprocessing
- Data visualization in Tableau
- Dashboard creation

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3. Users

The system is used by:

- Real estate analysts

- Marketing teams
- Company executives
- Property investors

These users interact with the dashboard to analyze housing trends.

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4. Output

The system produces:

- Interactive Tableau dashboards
- Housing market insights
- Visualization of property price trends

Process Explanation

1. Data Collection

Housing datasets are collected from sources such as:

- Kaggle
- Real estate databases
- CSV / Excel datasets

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2. Data Preparation

The dataset is cleaned and transformed.

Tasks include:

- Removing missing values
- Handling duplicates
- Converting data types
- Creating calculated fields

Example calculated fields:

- **House Age = Current Year – Year Built**
- **Years Since Renovation**
- **Price per Square Foot**

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3. Data Visualization

The prepared dataset is imported into **Tableau Desktop**.

Several charts are created, including:

- Bar charts
- Histograms
- Pie charts
- Scatter plots

These visualizations help analyze relationships between housing features and prices.

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4. Dashboard Development

All visualizations are combined into an **interactive Tableau dashboard**.

Features include:

- Filters
- Tooltips
- Interactive graphs

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5. Dashboard Publishing

The dashboard is published on:

- **Tableau Public**
  or
- **Tableau Server**

This generates an **embed link**.

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6. Web Integration

The dashboard is embedded into a **Flask web application**.

Flask allows users to access the dashboard through a browser.

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7. User Interaction

Users interact with the dashboard to analyze:

- Housing price trends
- Renovation impact on property value
- Distribution of house features

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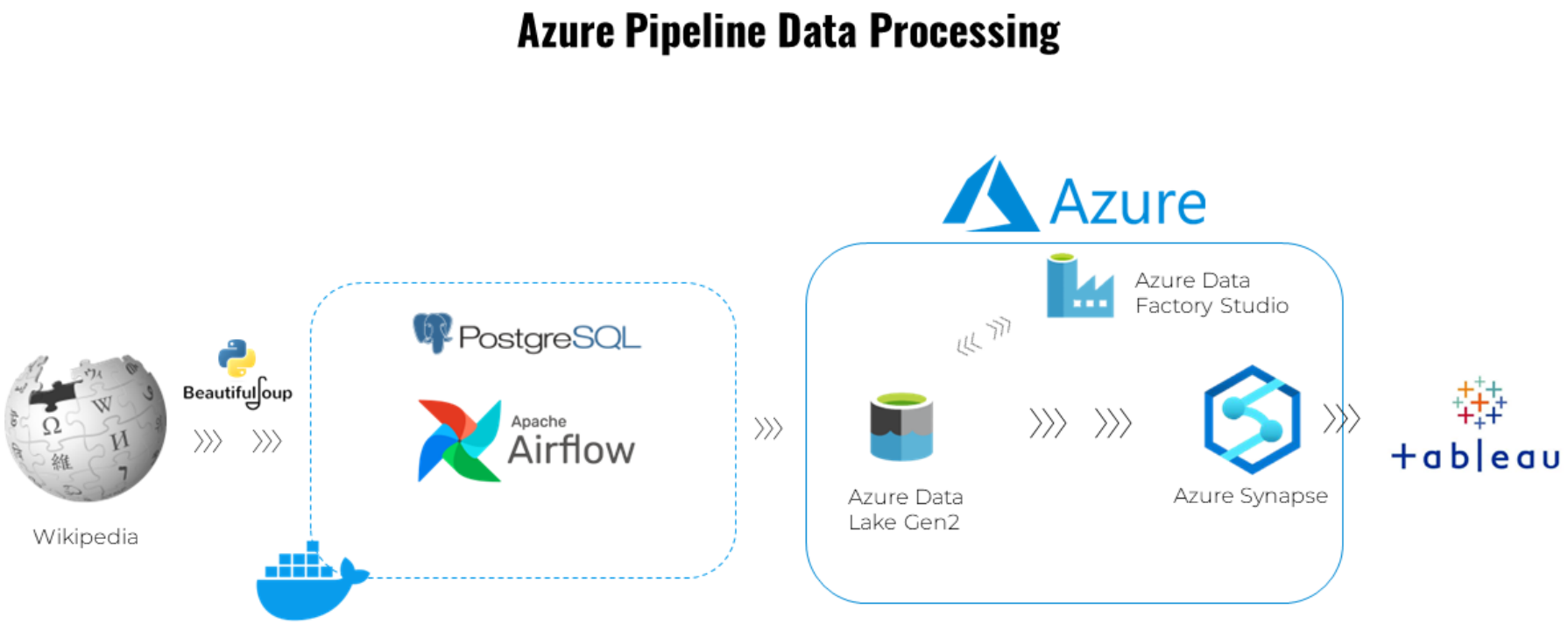
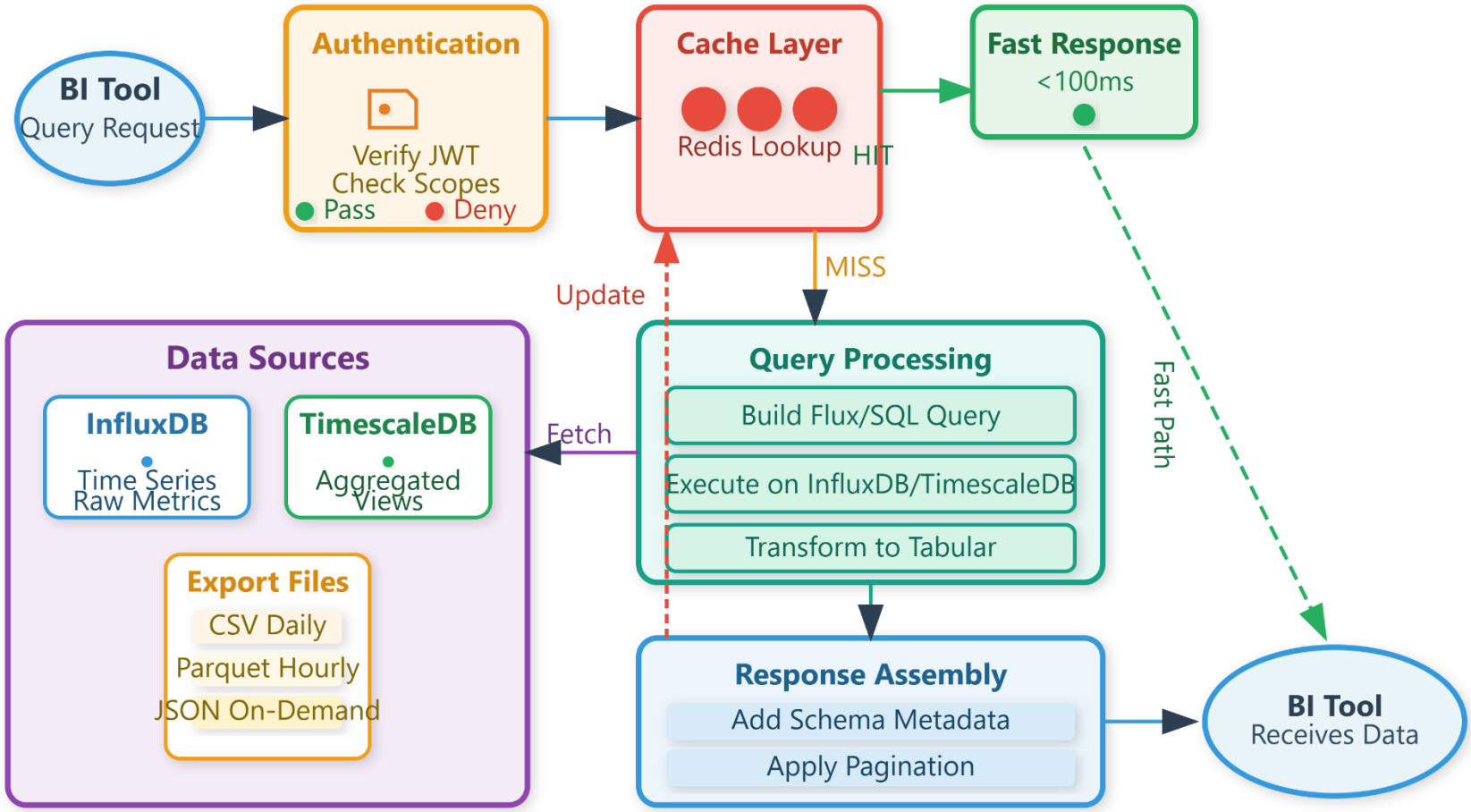
4. System Workflow Diagram



Rafael Vera

BI Data Flow & Processing Pipeline

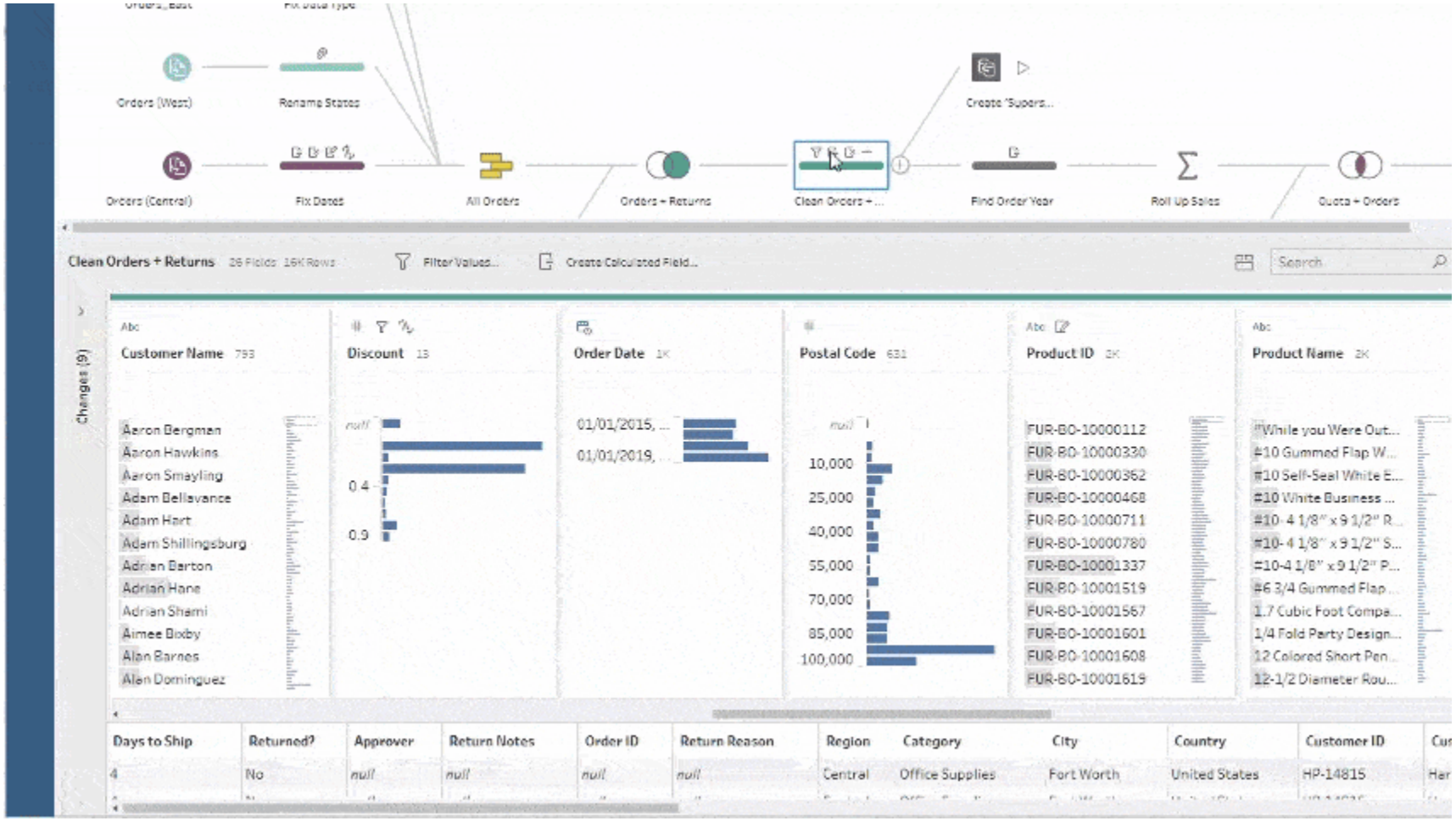


Key Transformations:

- Flux/SQL → JSON → Tabular Format
- Time Series → Aggregated Metrics
- Raw Fields → Business-Friendly Labels

Security Controls:

- OAuth 2.0 JWT Tokens
- Row-Level Data Filtering
- Scope-Based Permissions



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Workflow Steps

1. Collect housing dataset
2. Clean and preprocess data

3. Connect dataset to Tableau
4. Create visualizations
5. Develop interactive dashboard
6. Publish dashboard on Tableau Public
7. Integrate dashboard with Flask
8. Users access dashboard through web browser

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Real Estate Analyst	Renovation Analysis	USN-1	As an analyst, I want to view housing price trends	I can view interactive charts	High	Sprint-1
Real Estate Analyst	Feature Analysi	USN-2	As an analyst, I want to analyze renovation impact on house prices	Renovation vs price chart is visible	High	Sprint-1
Marketing Team	Trend Analysis	USN-3	As a marketing team member, I want to analyze housing features	I can compare bedrooms, bathrooms and floors	Medium	Sprint-2
Marketing Manager	Decision Support	USN-4	As a manager, I want to identify housing demand trends	Dashboard filters work correctly	Medium	Sprint-1
Executive	Login	USN-5	As an executive, I want summarized insights	I can see KPI summary charts	High	Sprint-1
Web User	Dashboard	USN-6	As a web user, I want to access the dashboard online	Dashboard loads in browser	High	Sprint-2
Administrator	Data Management	USN-7	As an admin, I want to update housing datasets	I can upload new datasets	Medium	Sprint-2